

DBMS Assignment – 3

Title : Implementation of SQL Functions

Description:

SQL Functions and its types

SQL provides many built-in functions to perform operations on data. These functions are useful while performing mathematical calculations, string concatenations, sub-strings etc.

Four types of SQL functions:

Aggregate Functions

MySQL supports the following aggregate functions:

Function	Description
AVG()	Returns the average of the values in the selected column
COUNT()	Returns the number of rows returned for a selection
MAX()	Returns the maximum value for a column
MIN()	Returns the minimum value of a column
SUM()	Returns the sum of the values in a specified column

Date Functions

Function	Description
<u>NOW()</u>	Returns the current date and time
<u>CURDATE()</u>	Returns the current date
<u>CURTIME()</u>	Returns the current time
<u>DATE()</u>	Extracts the date part of a date or date/time expression
<u>EXTRACT()</u>	Returns a single part of a date/time
<u>DATE_ADD()</u>	Adds a specified time interval to a date
<u>DATE_SUB()</u>	Subtracts a specified time interval from a date
<u>DATEDIFF()</u>	Returns the number of days between two dates
<u>DATE_FORMAT()</u>	Displays date/time data in different formats

Numeric Functions

Command	Query	Output
Abs(n)	Select abs(-15) from dual;	15
Ceil(n)	Select ceil(55.67) from dual;	56
Exp(n)	Select exp(4) from dual;	54.59
Floor(n)	Select floor(100.2) from dual;	100
Power(m,n)	Select power(4,2) from dual;	16
Mod(m,n)	Select mod(10,3) from dual;	1
Round(m,n)	Select round(100.256,2) from dual;	100.26
Trunc(m,n)	Select trunc(100.256,2) from dual;	100.23
Sqrt(m,n)	Select sqrt(16) from dual;	4

Char Functions

lower (char); upper (char);	<i>select lower ('HELLO') from dual;</i> <i>select upper ('hello') from dual;</i>	hello HELLO
ltrim (char,[set]);	<i>select ltrim ('cseit', 'cse') from dual;</i>	it
rtrim (char,[set]);	<i>select rtrim ('cseit', 'it') from dual;</i>	cse
replace (char,search string, replace string);	<i>select replace ('jack and jue', 'j', 'bl') from dual;</i>	black and blue
substr (char,m,n);	<i>select substr ('information', 3, 4) from dual;</i>	form

Assignment 3 Lab Questions

Perform following queries using SQL functions:

1. Find the cheapest book of SIBM library.

```
mysql> SELECT Bname, Price
-> FROM books
-> WHERE LibraryName = 'SIBM'
-> AND Price = (SELECT MIN(Price) FROM books WHERE LibraryName = 'SIBM');
+-----+-----+
| Bname                | Price |
+-----+-----+
| Database Management Systems | 950.00 |
+-----+-----+
1 row in set (0.00 sec)
```

2. Which library has the costliest book?

```
mysql> SELECT LibraryName
-> FROM books
-> WHERE Price = (SELECT MAX(Price) FROM books);
+-----+
| LibraryName |
+-----+
| SLS          |
+-----+
1 row in set (0.00 sec)
```

3. How many students from SIT issued the book?

```
mysql> SELECT COUNT(DISTINCT StudentName) AS SIT_Students_Count
-> FROM books
-> WHERE Institute = 'SIT' AND IssueDate IS NOT NULL;
+-----+
| SIT_Students_Count |
+-----+
| 4                  |
+-----+
1 row in set (0.01 sec)
```

4. What is the average cost of books in SITMN library?

```
mysql> SELECT AVG(Price) AS Average_Cost
-> FROM books
-> WHERE LibraryName = 'SITMN';
+-----+
| Average_Cost |
+-----+
| 999.500000   |
+-----+
1 row in set (0.00 sec)
```

5. What is the total cost of purchase made by SIT in the month of January to June?

```
mysql> SELECT SUM(Price) AS Total_Cost
-> FROM books
-> WHERE LibraryName = 'SIT'
-> AND MONTH(PurchaseDate) BETWEEN 1 AND 6;
+-----+
| Total_Cost |
+-----+
| 18250.00 |
+-----+
1 row in set (0.01 sec)
```

6. How many books are written by “Shruti”

```
mysql> SELECT COUNT(Bname) AS Books_By_Shruti
-> FROM books
-> WHERE Author = 'Shruti';
+-----+
| Books_By_Shruti |
+-----+
| 1 |
+-----+
1 row in set (0.00 sec)
```

7. What is the costliest book published by “Pragati Book Store”

```
mysql> SELECT Bname, Price
-> FROM books
-> WHERE Publisher = 'Pragati Book Store'
-> AND Price = (SELECT MAX(Price) FROM books WHERE Publisher = 'Pragati Book Store');
+-----+-----+
| Bname | Price |
+-----+-----+
| Advanced Artificial Intelligence | 12500.00 |
+-----+-----+
1 row in set (0.00 sec)
```

8. How many total copies of books do SIT has?

```
mysql> SELECT SUM(Copies) AS Total_SIT_Copies
-> FROM books
-> WHERE LibraryName = 'SIT';
+-----+
| Total_SIT_Copies |
+-----+
| 41 |
+-----+
1 row in set (0.00 sec)
```

9. What is the average cost of books written by “Shivam Kapoor”?

```
mysql> SELECT AVG(Price) AS Average_Cost
-> FROM books
-> WHERE Author = 'Shivam Kapoor';
+-----+
| Average_Cost |
+-----+
| 3649.500000 |
+-----+
1 row in set (0.00 sec)
```

10. How many books are sold by seller living in Pune?

```
mysql> SELECT COUNT(Bname) AS Books_From_Pune_Seller
-> FROM books
-> WHERE SellerCity = 'Pune';
+-----+
| Books_From_Pune_Seller |
+-----+
| 5 |
+-----+
1 row in set (0.00 sec)
```

11. Print the student name in capital who belongs to SSBS

```
mysql> SELECT UPPER(StudentName) AS Student_Name
-> FROM books
-> WHERE Institute = 'SSBS';
+-----+
| Student_Name |
+-----+
| AMIT |
+-----+
1 row in set (0.00 sec)
```

12. Add two months to the issue date of book written by “Shivam Kapoor”

```
mysql> SELECT Bname, IssueDate, DATE_ADD(IssueDate, INTERVAL 2 MONTH) AS Extended_Issue_Date
-> FROM books
-> WHERE Author = 'Shivam Kapoor';
+-----+-----+-----+
| Bname | IssueDate | Extended_Issue_Date |
+-----+-----+-----+
| Software Engineering | 2023-01-20 | 2023-03-20 |
| Cloud Computing Essentials | 2023-06-15 | 2023-08-15 |
+-----+-----+-----+
2 rows in set (0.01 sec)
```

13. What was the last day of the month when Satish issued the book?

```
mysql> SELECT LAST_DAY(IssueDate) AS Last_Day_Of_Issue_Month
-> FROM books
-> WHERE StudentName = 'Satish';
+-----+
| Last_Day_Of_Issue_Month |
+-----+
| 2023-08-31              |
+-----+
1 row in set (0.01 sec)
```

14. How many books are issued from January to march 2010 & 2020?

```
mysql> SELECT COUNT(Bname) AS Issued_Books
-> FROM books
-> WHERE YEAR(IssueDate) IN (2010, 2020)
-> AND MONTH(IssueDate) BETWEEN 1 AND 3;
+-----+
| Issued_Books |
+-----+
|             2 |
+-----+
1 row in set (0.01 sec)
```

15. How many books have copies less than 5 available in the SIBM library?

```
mysql> SELECT COUNT(Bname) AS Low_Stock_Books
-> FROM books
-> WHERE LibraryName = 'SIBM' AND Copies < 5;
+-----+
| Low_Stock_Books |
+-----+
|             2 |
+-----+
1 row in set (0.00 sec)
```

Conclusion :

I learned how to use built-in SQL functions like COUNT(), SUM(), and date modifiers to easily analyze and calculate data. Most importantly, I learned a crucial rule for using MIN() and MAX(): instead of putting them directly next to a column name, I must use a subquery to ensure the highest or lowest price accurately matches the correct book.